

Applications Information

Component Selection

MAX17335 has the following fixed components. These must remain the same in all designs. Voltage ratings listed are minimums. Higher ratings can be used, but should minimize derating on capacitors. The subscript denotes the pin name to which the component is connected. See the Typical Application Schematic for reference.

Table 12. MAX17335 Standard Components

COMPONENT	VALUE	MINIMUM RATING	NOTES
R _{BATT}	10Ω	—	Must match RC time with R _{PCKP} . Power rating should be sized with R _{ZVC} . If Zero-Volt charging is not used, use 1mA rating.
C _{BATT}	0.1μF	5V	Must match RC time with C _{PCKP} .
R _{PCKP}	1kΩ	—	Low Current. RC time matching is helpful for supplement mode detection.
C _{PCKP}	1nF	10V	This node can be exposed to higher voltage. A higher voltage capacitor can be used.
C _{CP}	0.1μF	10V	—
C _{REG2}	0.47μF	5V	—
R _{CHG}	5kΩ	—	—
C _{CHG}	22nF	—	—
R _{DIS}	1kΩ	—	Lower resistance can be used for faster DIS response. 0Ω is acceptable. Higher resistance increases FET Turn Off switching time.

Sense Resistor

Sense resistor selection is critical to MAX17335 operation. The charging and protection current range and resolution are defined by the sense resistor, as are the capacity range and resolution. [Table 13](#) provides guidance for regulation current ranges based on the sense resistor. A current range should be selected for the Constant Current regulation. Termination Current being below the charge current range is allowed. For managed DC-DC applications, the heat generated in the CHG/DIS FETs is significantly less, and smaller sense resistors and larger currents are allowed. For applications with fixed DC voltage input, and larger thermal budget, larger currents than the listed range are allowed, see [nIChgCfq1/2 Registers](#) for the maximum current range. For fixed DC applications, at low cell voltage, the MAX17335 can regulate in Constant Power mode until the cell voltage is high enough to enter CC mode. See [Table 19](#) for heat limits.

Table 13. Sense Resistor Selection

CHARGE SOURCE	R _{SENSE} (mΩ)	CHARGE CURRENT RANGE (mA)
Managed DC-DC or Switching Charger	2	1850—17500
Managed DC-DC or Switching Charger	5	740—7000
Managed DC-DC or Switching Charger	10	185—3500
5V or USB	10	185—2331
5V or USB	20	92.5—1166
5V or USB	50	37—466
5V or USB	100	18.5—233
5V or USB	200	9.25—117
5V or USB	500	3.7—47